



Cape Peninsula
University of Technology

FACULTY OF INFORMATICS & DESIGN

ADVANCED DIPLOMA in INFORMATION & COMMUNICATIONS TECHNOLOGY

Subject Guide

APPLICATIONS DEVELOPMENT THEORY 4

SUBJECT GUIDE FOR:	2020
COURSE CODE:	ADT470S
NQF LEVEL:	7
PRE-REQUISITE SUBJECTS:	Applications Development Theory 3 or equivalent

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1. INTRODUCTION

Welcome to Applications Development Theory 4. This Study Guide gives an overview of this course, the objectives, expected learning outcomes, and the mode of assessment that will be engaged in the administration of this course. The course outline, relevant textbooks, other resources are also included. It also offers a reference to important dates and events relating to this course.

2. GENERAL INFORMATION

2.1 Contact Information

DESIGNATION	NAME	LOCATION	TELEPHONE NUMBER	E-MAIL ADDRESS
Subject Lecturer	G Khan	Engineering 2.8	(021) 460 3455	khang@cput.ac.za
Domain Coordinator	T Makhurane	Engineering 2.11	(021) 460-3348	makhuranet@cput.ac.za
Head of Department	B. Kabaso	Engineering 2.30	(021) 460-3713	kabasob@cput.ac.za
Secretary	N. Allie	Engineering 2.1	(021) 460-3010	alien@cput.ac.za

2.2 Class Times and Venues

As published on <http://myclassroom.cput.ac.za> and notice boards. This subject consists of one period per week.

2.3 Learner Management System (LMS)

Blackboard used for general communications and announcements, posting of course material, assignment and announcements. It is the responsibility of the learner to check Blackboard regularly at <https://myclassroom.cput.ac.za/>

2.4 Subject Credits

This subject has 15 credits at level 7 on the NQF. This means that the average student will take 150 hours per semester to master the outcomes of the subject. Of this 30% of the time will be contact hours and the remainder will be self-study and other tasks, assignments and tests. Please refer to the Faculty prospectus on <http://www.cput.ac.za/academic/faculties/informaticsdg/prospectus> for further information.

2.4.1 Length of Course

METHOD	NOTIONAL HOURS (1 Notional hour = 10 hours)
Formal Lectures	
Practical / Assignments	
Self-Study	
Other	
TOTAL	

2.5 Ethical Conduct

Conduct in this subject is governed by CPUT's guidelines on the following ethical matters :-

TOPIC	CPUT REFERENCE DOCUMENT AVAILABLE AT
Roles and Responsibilities	CPUT's Policy Docs and Student Guidelines
	https://www.cput.ac.za/storage/study_at_cput/Student_Rules%20Regulations_2018.pdf
Class attendance and participation r	page 18 of CPUT's General Handbook Academic and Student Rules and Regulations
	https://www.cput.ac.za/storage/study_at_cput/Student_Rules%20Regulations_2018.pdf
Grievance procedures	page 61 of CPUT's General Handbook Academic and Student Rules and Regulations:
	https://www.cput.ac.za/storage/study_at_cput/Student_Rules%20Regulations_2018.pdf
Plagiarism	page 30 of CPUT's General Handbook Academic and Student Rules and Regulations:
	https://www.cput.ac.za/storage/study_at_cput/Student_Rules%20Regulations_2018.pdf

*NB: An amended version of the General Handbook may be available on the university website.

2.6 Recognition of Prior Learning (RPL)

In a situation where there is a claim of previous learning that equates to a qualification of Information Technology, the recognition of such qualification will be at the discretion of the higher education institution. A structured process for the assessment of individual learners will be followed, vis-a-vis the exit level outcomes of the qualification on a case-by-case basis. Independent assessors will moderate this process and the assessment of individual cases.

In this course, we recognize RPL by granting a credit to learners who demonstrate both a theoretical and practical knowledge of software engineering and develop large software intensive systems. This is not to be confused with an exemption granted for a similar subject studied at another recognized tertiary institution.

3. COURSE INFORMATION

3.1 Overview of the Course

The need for automation in order to promote efficiency and effectiveness of human operations in diverse fields of application cannot be over-emphasized. The concept of software engineering enables the design and development of software intensive systems based on sound engineering principles. This course is designed to provide understanding of some of the modern concepts of software engineering and their application to solving industrial scale problems. It will provide an opportunity for students to

work together in groups in clusters that model software project teams in order to deliver some project objectives based on assigned tasks.

3.2 Course Objectives

- i. To provide students with an understanding of modern software engineering concepts and their application to industrial scale problems;
- ii. To provide students with the knowledge of software design and modelling as core activities of the software development process;
- iii. To study a myriad of tools and technologies that are relevant to different software process models and soft implementation approaches; and
- iv. To enable students with the capability to design and develop usable software intensive systems based on the application of modern software engineering concepts, tools and techniques.

3.3. Lecture Delivery Methods

The mode of lecture delivery will be based on the following:

□ Lectures , □ Mandatory Reading List and Assigned Book readings, □ Paper Reviews, □ Software Development projects (Group)

3.4. Textbooks

The following textbooks are relevant to this course:

- i) *Software Engineering*, by Ian Sommerville 10th edition, Pearson Education Limited, 2016.
- ii) *Pressman, R.S. Software Engineering: A Practitioner's Approach. Adapted by D. Ince. (London: McGraw Hill, 2010), seventh edition [ISBN 9780073375977].*

3.5. Library and Readings

Additional reading is an essential part of the course and you are expected to make use of the library and the Internet

4. ASSESSMENT

4.1 Assessment policy and regulations

This subject has four assessments. Assessments are a combination of Blackboard online tests, iLearning quizzes and assignments. Assignments are administered via Blackboard.

Test queries: Last day for any queries regarding a test is one week after the test was discussed in class. Thereafter the test mark cannot be changed.

Absence from a test: If you are absent from a test, a valid medical certificate must be given to your lecturer (via the Secretary) within one week after the test was written. You must also inform your lecturer on the day of the test that you are ill.

Due dates must be strictly adhered to. NO late submissions are allowed for any subject in this Advanced Diploma.

4.2 Assessment Opportunities:

There will be one test per term. At the end of the year there will be an Exception/Sick test for the semester.

	ASSESSMENT BREAKDOWN AND WEIGHTS				
ASSESSMENT CRITERIA	1st TERM	2nd TERM	3rd TERM	4th TERM	TOTALS
Nature of assessment	Written Test	Written Test & Assignments	Written Test	Assignments	
Will assessment be included in this term?	Yes	Yes	Yes	Yes	100%
Total assessment weight for the term	20%	30%	30%	20%	
Will the assessment be moderated?	No	Yes	Yes	No	

4.3 Tests and Assignment Guidelines

For a group work, a maximum of 4 students per group will be allowed. Group work should be done in a way that encourages individual contributions. Deadlines for submission of assignments must be adhered to in order to qualify for the maximum score.

4.3.1 Assignments

□ Assignments on software design and development will target specific learning outcomes and are detailed in the Assignment document on Blackboard.

5 SUBJECT INFORMATION

5.1 Subject Outcomes

At the end of this course, students will be able to:

OUTCOME NO	OUTCOME DESCRIPTION
SO1	Understand the generic methods for system modeling at various levels of abstraction such as requirements, architecture and design
SO2	Apply modern RE methods such as goal-oriented requirement engineering, Agile requirements engineering, and requirement boilerplate to identify and model requirements
SO3	Understand quality requirements issues
SO4	Engage state-of-the art techniques to ensure quality assurance and build dependable systems
SO5	Understand state of the practice agile methods application and concepts such as DevOps, continuous integration, and model-driven development
SO6	Apply learned concepts to solve industrial scale problems by navigating several aspects of the software process lifecycle
SO7	Explain key issues and solutions of managing complex software-intensive systems

5.2 Critical Cross-Field Outcomes (CCFO's)

The Critical Cross-Field Outcomes to be addressed by this course are problem solving, group work, organization and management, application, evaluation, communication, presentation, and research.

Students will have the opportunity to enhance their problem solving skills through the acquisition of theoretical concepts that underline database design, and application of the knowledge gained to solve the real world problems. Group work and communication are addressed by collaborative work on solving the database problems.

5.3 Topic Organisation

No	Topic	Week of
Term 1	THEME 1: SOFTWARE REQUIREMENTS and DESIGN	
1	Requirements specification module: focuses on methods to transit from user requirements to high-quality technical requirements.	12/2/2020
2	Fundamental design principles	19/2/2020
3	Design in Agile	26/2/2020
4	Plan-driven and Open Source projects	11/3/2020
5	Assignment 1 and Exam on Term 1 work	11/3/2020
Term 2	THEME 2: SOFTWARE DESIGN	
6	Architecture Patterns, Design patterns	8/4/2020
7	Software Architectures, Design in different applications/technologies, SOA	15/4/2020
8	Model Driven Development	22/4/2020
9	Testing and quality assurance model focuses on testing, code review, and verification and validation strategies and methods, especially for high-assurance systems.	29/4/2020
10	Software Quality and Evolution	13/5/2020
	Assignment and Exam on Semester 1 work	
Term 4	THEME 3: INTENSIVE SYSTEMS	
11	Process improvement module: focuses on modern agile methods, e.g. DevOps, DevSecOps. Process and Product Metrics	22/7/2020
12	Software Dependability: will focus of development of secure, reliable, safe and resilient systems	29/7/2020

13	Software reuse	5/8/2020
14	Dependence Engineering	19/8/2020
	Assignment and Exam on Term 3 work	
Term 4	THEME 4: SOCIOTECHNICAL SYSTEMS	
15	Sociotechnical systems, Systems of systems	9/10/2020
	Assignment	

6. STUDENT ACKNOWLEDGEMENT

CAPE PENINSULA UNIVERSITY OF TECHNOLOGY

STUDENT ACKNOWLEDGEMENT

2020

ADVANCED DIPLOMA: INFORMATION and COMMUNICATIONS TECHNOLOGY

APPLICATIONS DEVELOPMENT THEORY 4

ACKNOWLEDGE THAT YOU HAVE READ THE SUBJECT GUIDE AND UNDERSTAND ITS CONTENT

Please read the following statement and complete the spaces that follow. Make a copy of this page for your own reference. Hand-in this original page during the next class session.

I have read the subject guide of **APPLICATIONS DEVELOPMENT THEORY 4** and understand the information and my responsibilities as a learner for this subject. I am also aware that the Subject Guide may be revised due to unforeseen circumstances and it remains my responsibility to adhere to the most recent guide.

Name & Surname: (Print)

Student Number:

Contact number: (home/res) Contact number: (cell)

Signature:

Date: